

# Giorgos Thanasoulis

## Electrical & Computer Engineer



### Communication

Thessaloniki, 54642  
geor.thanasoulis@gmail.com  
+30 698 5702 189  
*LinkedIn Profile:*  
Giorgos Thanasoulis

Electrical and Computer Engineer with a strong interest in Artificial Intelligence and its real-world applications. Currently involved in two European research projects, contributing to innovative solutions in data analysis and intelligent systems. My academic background includes work on image-based defect detection, while my professional experience has strengthened my problem-solving, critical thinking, and collaboration skills. Passionate about continuous learning and applying technology to address complex challenges.

### Fulfilled Military Obligations

*(November 2023 – August 2024)*

### Education

Democritus  
University of Trace  
Xanthi

*October 2018 – December 2024*

#### Electrical and Computer Engineer

- Electronics and Information Systems
- Integrated Master (300 ECTS)
- Diploma Thesis titled “Imperfection Detection from Photovoltaic-Panel Electroluminescence Images”.

An automated system for defect detection and categorization of photovoltaic panels was developed using deep learning. Four datasets were created and used to train Custom Convolutional Neural Networks (CNNs). Furthermore, Pretrained models such as VGG-16, VGG-19, ResNet-18, and ResNet-34 were also utilized through transfer learning. The results of these models were compared with the CNNs' outcomes, evaluating their accuracy and efficiency.

### Experience

*February 2025 – Present*

#### Data Scientist – Centre for Research & Technology Hellas (CERTH)

Currently, participating in two European research projects focused on the design of AI algorithms for forecasting and optimizing resource orchestration in cloud and edge computing environments, as well as the development of secure data processing methods for trustworthy and privacy-preserving artificial intelligence models.

Responsibilities include:

- Designing and implementing AI forecasting models optimizing resource orchestration in cloud and edge computing environments.

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|                       | <ul style="list-style-type: none"> <li>Developing synthetic data generation techniques, particularly using advanced deep learning architectures, serving as supplementary data for augmentation.</li> </ul>                                                                                                                                     |
| <b>Skills</b>         | <ul style="list-style-type: none"> <li>Microsoft Excel, Word, Latex &amp; Power Point</li> <li>Python, Pytorch, Tensorflow</li> <li>Machine Learning, Deep Learning</li> <li>Scikit Learn</li> <li>Docker</li> <li>Git, Github</li> <li>Linux</li> <li>Continuous knowledge enhancement</li> <li>Critical Thinking</li> <li>Teamwork</li> </ul> |
| <b>Certifications</b> | <p>2023 - 2024</p> <ul style="list-style-type: none"> <li>Python for Data Science and Machine Learning Bootcamp - <i>Udemy</i></li> <li>The Complete Neural Networks Bootcamp: Theory, Applications - <i>Udemy</i></li> <li>The Complete SQL Bootcamp: Go from zero to Hero - <i>Udemy</i></li> </ul>                                           |
| <b>Languages</b>      | <ul style="list-style-type: none"> <li>Greece (Native)</li> <li>English (Level B2)</li> <li>In Progress: Preparation for Proficiency</li> </ul>                                                                                                                                                                                                 |